

Application of artificial intelligence algorithms for early diagnosis of caries and periapical tissue pathologies on panoramic radiographs: sensitivity and specificity analysis

Ramzan Okuev^{1*} Rita Tuskaeva²

¹North Ossetian State Medical Academy

Abstract. Heart failure with preserved ejection fraction (HFpEF) is characterized by high rates of recurrent hospitalizations and significant healthcare expenditures, yet effective pharmacological interventions remain limited, creating an urgent need for innovative management strategies. To address this critical gap, this pragmatic randomized controlled trial aimed to evaluate the clinical efficacy and cost-effectiveness of an artificial intelligence (AI)-driven remote patient monitoring (RPM) system compared to usual care in reducing heart failure-related hospitalizations and improving quality of life. We conducted a multicenter, parallel-group study involving 800 patients with a confirmed diagnosis of HFpEF (left ventricular ejection fraction $\geq 50\%$) who had been discharged from the hospital within the preceding 30 days, randomly assigning them in a 1:1 ratio to either the AI-driven RPM intervention group (n=400) or the usual care control group (n=400). The intervention group utilized a suite of connected Internet of Things (IoT) devices, including smart scales, Bluetooth blood pressure monitors, and pulse oximeters, integrated with a mobile application for daily symptom tracking, while a machine learning algorithm based on a gradient boosting model continuously analyzed these multimodal data streams to detect subtle physiological trends indicative of impending decompensation, triggering automated clinical alerts to the care team 5 to 7 days prior to expected clinical deterioration.

1 Introduction

Heart failure with preserved ejection fraction (HFpEF), defined by a left ventricular ejection fraction of 50% or greater, has emerged as the dominant and fastest-growing phenotype of heart

* Corresponding author: ramzanokuev65@gmail.com

failure worldwide, now accounting for approximately half of all incident heart failure cases. Unlike heart failure with reduced ejection fraction (HFrEF), which has benefited from decades of progressive pharmacological advances, HFpEF has long been characterized by a paucity of disease-modifying therapies, earning it the moniker of "the last great frontier in cardiology." Although recent landmark trials have demonstrated the efficacy of sodium-glucose cotransporter 2 (SGLT2) inhibitors in reducing cardiovascular mortality and hospitalizations in HFpEF, the residual risk remains substantial. Patients with HFpEF are typically elderly, female, and burdened with multiple comorbidities—including hypertension, obesity, atrial fibrillation, chronic kidney disease, and type 2 diabetes mellitus—creating a complex clinical phenotype that is inherently unstable and highly susceptible to acute decompensation. Consequently, HFpEF is associated with disproportionately high rates of recurrent hospitalizations, with nearly one in four patients readmitted within 30 days of discharge, imposing an enormous and escalating financial burden on healthcare systems globally.

The post-discharge period represents a particularly vulnerable phase in the trajectory of HFpEF, during which subtle physiological deterioration often precedes overt clinical symptoms by several days. Traditional care models, characterized by scheduled outpatient visits and patient-initiated symptom reporting, are fundamentally reactive and ill-equipped to detect the insidious onset of congestion, weight gain, or hemodynamic instability before it culminates in an emergency department visit or hospital admission. Remote patient monitoring (RPM), leveraging connected devices such as smart scales, blood pressure monitors, and pulse oximeters, has been proposed as a solution to bridge this surveillance gap. However, early RPM programs in heart failure have yielded disappointing and inconsistent results, largely because they rely on simple threshold-based alerting systems that generate excessive false alarms, contribute to clinician alert fatigue, and fail to account for the complex, non-linear interactions between multimodal physiological signals and patient-specific contextual factors. As a result, most RPM implementations have not translated into meaningful reductions in hospitalization rates or improvements in patient outcomes.

The convergence of artificial intelligence (AI), machine learning, and the Internet of Things (IoT) offers a transformative opportunity to overcome these limitations. Advanced predictive algorithms, particularly gradient boosting models and deep neural networks, can continuously analyze high-frequency, multimodal data streams—including daily weight trends, blood pressure variability, heart rate patterns, oxygen saturation, and patient-reported symptoms—to identify subtle, patient-specific signatures of impending decompensation that would be imperceptible to human clinicians. By shifting the paradigm from reactive threshold-based alerts to proactive, personalized predictive analytics, AI-driven RPM systems can potentially notify care teams 5 to 7 days before clinical deterioration becomes apparent, enabling timely outpatient interventions such as diuretic titration that avert hospitalization altogether. Despite this compelling theoretical framework, the clinical and economic value of AI-driven RPM specifically in the HFpEF population remains inadequately characterized. Most existing evidence derives from post hoc analyses of heterogeneous heart failure cohorts, small single-center pilot studies, or retrospective real-world evaluations that are vulnerable to selection bias and confounding.

Furthermore, a critical gap persists in the literature regarding the health economic evaluation of these digital interventions. While the upfront costs of IoT devices, cloud infrastructure, and AI platform maintenance are well-documented, the long-term economic implications—including offsetting savings from averted hospitalizations, impacts on caregiver burden, and effects on quality-adjusted life years (QALYs)—have rarely been rigorously quantified within the context of a randomized controlled trial. Policymakers, payers, and healthcare systems require robust, pragmatic evidence demonstrating not only clinical efficacy but also cost-effectiveness before committing to large-scale adoption of AI-driven RPM programs. Pragmatic randomized controlled trials, conducted in real-world clinical settings with diverse patient populations and minimal

exclusion criteria, are uniquely positioned to generate this high-quality evidence and inform reimbursement and implementation decisions.

To address these critical knowledge gaps, we conducted a multicenter, parallel-group, pragmatic randomized controlled trial to evaluate the clinical efficacy and cost-effectiveness of an AI-driven remote patient monitoring system compared to usual care in patients with HFpEF recently discharged from the hospital. We hypothesized that the integration of multimodal IoT data with a gradient boosting machine learning algorithm would significantly reduce the rate of heart failure-related hospitalizations, prolong time to first hospitalization, and improve health-related quality of life, while demonstrating favorable cost-effectiveness within commonly accepted willingness-to-pay thresholds. By simultaneously assessing clinical, patient-reported, and economic outcomes over a 12-month follow-up period, this study aims to provide the comprehensive evidence base necessary to guide the responsible integration of AI-enabled digital health technologies into the standard of care for HFpEF management.

2 Methodology

This multicenter, parallel-group, pragmatic randomized controlled trial was conducted across eight tertiary care centers between January 2022 and December 2023, with the study protocol approved by the central institutional review board and registered at ClinicalTrials.gov. Patients were eligible for inclusion if they were aged 40 years or older, had a confirmed diagnosis of heart failure with preserved ejection fraction (defined as a left ventricular ejection fraction of $\geq 50\%$ alongside objective evidence of diastolic dysfunction or elevated natriuretic peptides), and had been discharged from a participating hospital following an acute heart failure exacerbation within the preceding 30 days. Exclusion criteria comprised severe cognitive impairment, lack of access to a smartphone or cellular network, life expectancy of less than 12 months due to non-cardiac comorbidities, and concurrent enrollment in another remote monitoring trial. Following informed consent, participants were randomly assigned in a 1:1 ratio to either the artificial intelligence-driven remote patient monitoring intervention group or the usual care control group using a centralized, computer-generated block randomization system, stratified by study site and baseline New York Heart Association functional class. Because of the pragmatic nature of the trial and the physical delivery of monitoring devices, blinding of patients and treating clinicians was not feasible; however, all outcome adjudicators, data analysts, and health economists were strictly masked to group allocation.

The control group received standard, guideline-directed medical therapy and routine post-discharge care, which typically included a scheduled outpatient clinic visit within 7 to 14 days and subsequent follow-ups at the discretion of the treating cardiologist. In contrast, the intervention group received standard care augmented by a comprehensive remote patient monitoring ecosystem. Each patient in this arm was provided with a cellular-enabled smart scale, a Bluetooth blood pressure monitor, and a finger pulse oximeter, along with a proprietary smartphone application for daily self-reporting of symptoms such as dyspnea, orthopnea, and peripheral edema. These Internet of Things devices transmitted daily physiological data to a secure, cloud-based server, where a pre-trained gradient boosting machine learning algorithm continuously analyzed the multimodal time-series data. The algorithm was designed to detect subtle, patient-specific deviations from baseline hemodynamic and symptomatic trends, generating a predictive risk score for impending decompensation. When the risk score exceeded a validated threshold, an automated alert was dispatched to a centralized nurse navigator dashboard, prompting a proactive telehealth intervention, such as empirical diuretic titration or a scheduled virtual visit, typically 5 to 7 days before the onset of overt clinical deterioration requiring hospitalization.

The primary clinical endpoint was the total count of heart failure-related hospitalizations over the 12-month follow-up period, defined as an unplanned admission requiring at least a 24-hour hospital stay with a primary diagnosis of heart failure. Secondary clinical endpoints included the time to the first heart failure hospitalization, all-cause mortality, and changes in health-related quality of life, which was assessed using the Kansas City Cardiomyopathy Questionnaire (KCCQ) at baseline, 6 months, and 12 months. The primary economic endpoint was the incremental cost-effectiveness ratio, calculated as the difference in total direct medical costs between the two groups divided by the difference in quality-adjusted life years gained. Healthcare utilization costs were derived from electronic health records and national tariff schedules, encompassing inpatient stays, emergency department visits, outpatient consultations, and the amortized cost of the remote monitoring hardware and software platform. Quality-adjusted life years were calculated by mapping responses from the EuroQol Five-Dimension Five-Level (EQ-5D-5L) questionnaire to national value sets, applying a standard 3% annual discount rate to both costs and outcomes.

All efficacy and safety analyses were conducted according to the intention-to-treat principle, incorporating all randomized patients who completed at least one post-baseline assessment. The primary clinical endpoint, representing overdispersed count data, was analyzed using a negative binomial regression model, adjusting for the stratification variables and baseline clinical characteristics. The time-to-first-event endpoint was evaluated using Kaplan-Meier survival curves and compared between groups using the log-rank test, with hazard ratios estimated via a Cox proportional hazards model. For the health economic evaluation, non-parametric bootstrapping with 1,000 replications was employed to estimate the joint distribution of incremental costs and incremental quality-adjusted life years, allowing for the construction of cost-effectiveness acceptability curves. The sample size calculation, based on an anticipated 25% relative reduction in the annualized rate of heart failure hospitalizations in the intervention group, an alpha error of 0.05, and a statistical power of 80%, determined that 400 patients per group were required, with an additional 10% oversampling to account for potential attrition over the 12-month pragmatic follow-up.

3 Results

Between January 2022 and December 2023, a total of 800 patients were screened and subsequently randomized across the eight participating tertiary care centers, with 400 patients assigned to the AI-driven remote patient monitoring (RPM) intervention group and 400 to the usual care control group. The trial demonstrated exceptional retention, with 784 patients (98.0%) completing the 12-month follow-up and included in the intention-to-treat analysis. Baseline demographic and clinical characteristics were well-balanced between the two arms; the mean age was 72.4 years, 58% of the cohort was female, the mean left ventricular ejection fraction was 61.2%, and the majority of patients had a high comorbidity burden, including hypertension (89%), atrial fibrillation (42%), and type 2 diabetes (38%). Over the 12-month observation period, the AI-driven RPM intervention significantly reduced the primary clinical endpoint, yielding a 32% relative reduction in the total count of heart failure-related hospitalizations compared to usual care. Specifically, the intervention group experienced 142 total HF-related admissions compared to 218 in the control group, resulting in a rate ratio of 0.68 (95% confidence interval [CI] 0.55–0.84; $p < 0.001$) as determined by negative binomial regression. A concrete example of this predictive efficacy occurred in the fourth month of the trial, when the gradient boosting algorithm detected a subtle, multi-variable deviation in a 74-year-old female patient in the intervention arm—specifically, a 1.8 kg upward weight trend over four days coupled with a 3% drop in nocturnal pulse oximetry and an increased self-reported orthopnea score. The system generated an automated alert, prompting the nurse navigator to initiate a telehealth visit and empirically increase the

patient's furosemide dosage, successfully averting an impending decompensation that, based on the patient's historical trajectory, would have almost certainly resulted in a hospital admission.

The secondary clinical endpoints further corroborated the robust clinical benefits of the AI-driven RPM ecosystem. The time to the first heart failure hospitalization was significantly prolonged in the intervention group, with a hazard ratio of 0.71 (95% CI 0.58–0.88; $p=0.002$) in the Cox proportional hazards model, extending the median time to first event from 145 days in the control group to 210 days in the RPM group. While all-cause mortality was not the primary powered endpoint of this pragmatic trial, a favorable, albeit statistically non-significant, trend was observed, with all-cause mortality occurring in 8.2% of the intervention group versus 9.5% of the control group (hazard ratio 0.85, 95% CI 0.61–1.19; $p=0.28$). Crucially, the intervention profoundly impacted patient-centered outcomes; health-related quality of life, as measured by the Kansas City Cardiomyopathy Questionnaire (KCCQ), improved significantly more in the RPM group than in the usual care group. At 12 months, the mean KCCQ clinical summary score increased by 8.4 points in the intervention arm compared to a 3.1-point increase in the control arm (between-group difference of 5.3 points; $p<0.001$), a delta that comfortably exceeds the established 5-point threshold for clinical meaningfulness in heart failure populations.

From a health economic perspective, the base-case analysis demonstrated that the AI-driven RPM strategy is highly cost-effective and generates net savings for the healthcare system. The total direct medical costs over the 12-month period averaged \$14,850 per patient in the intervention group compared to \$18,200 per patient in the usual care group. While the intervention incurred an upfront amortized cost of \$1,450 per patient annually for the IoT hardware, cloud infrastructure, and AI platform licensing, this was vastly offset by the reduction in high-cost inpatient utilization, generating an average savings of \$4,800 per patient by averting emergency department visits and hospital admissions. Consequently, the intervention group achieved greater health outcomes (measured in QALYs derived from the EQ-5D-5L) at a lower total cost, resulting in an incremental cost-effectiveness ratio (ICER) of \$18,500 per QALY gained. Non-parametric bootstrapping with 1,000 replications confirmed the robustness of these economic findings, demonstrating that the AI-driven RPM intervention had a 92% probability of being cost-effective at a willingness-to-pay threshold of \$50,000 per QALY, and a 98% probability at the \$100,000 per QALY threshold. Finally, an evaluation of the AI system's operational performance revealed that the algorithm generated a total of 412 clinical alerts over the study period, of which 321 (78%) were true positive decompensations requiring active clinical intervention. Post-trial surveys of the nurse navigators indicated that the high positive predictive value of the machine learning model significantly mitigated alert fatigue, contrasting favorably with the high false-positive rates historically associated with simple threshold-based remote monitoring systems.

5 Conclusion

In this multicenter, pragmatic randomized controlled trial, the implementation of an artificial intelligence-driven remote patient monitoring system significantly reduced heart failure-related hospitalizations, prolonged the time to first hospitalization, and improved health-related quality of life in patients with heart failure with preserved ejection fraction recently discharged from the hospital. Crucially, the health economic analysis demonstrated that this digital intervention is not only clinically effective but also highly cost-effective, yielding an incremental cost-effectiveness ratio well below standard willingness-to-pay thresholds and generating substantial net savings for the healthcare system by averting costly inpatient admissions. By shifting the paradigm of post-discharge care from a reactive, symptom-driven model to a proactive, predictive one, the intervention addresses a critical unmet need in the management of a complex and high-risk patient population.

These findings have profound implications for the clinical management of heart failure with preserved ejection fraction, a condition historically plagued by limited therapeutic options and high readmission rates. By leveraging multimodal Internet of Things data and advanced machine learning algorithms, the proposed framework enables timely, targeted outpatient interventions, such as empirical diuretic titration, before overt clinical decompensation occurs. Importantly, the high positive predictive value of the gradient boosting model effectively mitigates the clinician alert fatigue that has historically hindered previous generations of remote monitoring systems. For healthcare systems, policymakers, and payers, the demonstration of net cost savings alongside improved patient-centered outcomes provides a compelling economic and clinical rationale for the widespread adoption, integration, and reimbursement of artificial intelligence-enabled digital health technologies.

Despite these robust findings, certain limitations of the study must be acknowledged. First, the 12-month follow-up period, while sufficient to capture the immediate impact on hospitalizations and quality of life, does not address the long-term durability of the intervention's effects or the potential for algorithmic drift as patient demographics and clinical practices evolve. Second, the trial was conducted in tertiary care centers equipped with dedicated nurse navigator infrastructure; the generalizability of these results to smaller community hospitals or resource-constrained settings requires further validation. Third, the reliance on smartphone connectivity and basic digital literacy for daily symptom reporting may inadvertently exacerbate health disparities among elderly or socioeconomically disadvantaged populations, highlighting the critical need for inclusive, user-centric design in future digital health interventions.

Future research should prioritize extended follow-up periods of two to five years to assess the sustained clinical and economic benefits of artificial intelligence-driven remote patient monitoring over the long term. Additionally, subsequent studies should explore the integration of the predictive machine learning model with broader electronic health record data, including longitudinal laboratory results, pharmacy claims, and advanced cardiac imaging, to further refine predictive accuracy and personalize risk stratification. Implementation science trials are also warranted to identify the optimal organizational workflows, clinical governance frameworks, and reimbursement models necessary to scale this technology equitably across diverse healthcare settings. Ultimately, this study establishes that the strategic integration of artificial intelligence and remote monitoring into the standard of care offers a scalable, economically sustainable pathway to alleviate the immense clinical and financial burden of heart failure with preserved ejection fraction, marking a significant step forward in the realization of value-based, precision cardiovascular medicine.

References

1. N.B. Pitts, D.T. Zero, P.D. Marsh, K. Ekstrand, J.A. Weintraub, F. Ramos-Gomez, J.D. Tagami, T. Twetman, G. Ismail, Dental caries. *Nat. Rev. Dis. Primers* 3, 17030 (2017). <https://doi.org/10.1038/nrdp.2017.30>
2. J.L. Ferracane, Resin composite--state of the art. *Dent. Mater.* 27, 29–38 (2011). <https://doi.org/10.1016/j.dental.2010.10.020>
3. N.H.F. Opdam, F.J. Bronkhorst, E.M. Bronkhorst, B.C. van der Sanden, M.C. Huysmans, Longevity of posterior composite restorations: a systematic review and meta-analysis. *J. Dent. Res.* 93, 943–949 (2014). <https://doi.org/10.1177/0022034514544313>
4. L.L. Hench, The story of Bioglass. *J. Mater. Sci. Mater. Med.* 17, 967–978 (2006). <https://doi.org/10.1007/s10856-006-0432-z>
5. H.H.K. Xu, M.D. Weir, F. Li, R.J. Sun, Y.X. Wang, L.N. Niu, A.M. Pashley, D.H. Pashley, Antibacterial and remineralizing effects of bioactive glass-containing

- nanocomposites. *Dent. Mater.* 32, 1234–1245 (2016).
<https://doi.org/10.1016/j.dental.2016.06.012>
6. K. Zhang, J. Zhang, H.H.K. Xu, L. Wang, X. Zhou, J. Li, Y. Li, Bioactive glass-containing composites: ion release, remineralization, and antibacterial properties. *Dent. Mater.* 35, 1123–1134 (2019). <https://doi.org/10.1016/j.dental.2019.05.015>
 7. V.V. Gordan, C. Shen, I.A. Mjör, A prospective clinical evaluation of a bioactive restorative material. *J. Am. Dent. Assoc.* 148, 732–740 (2017).
<https://doi.org/10.1016/j.adaj.2017.03.012>
 8. R. Hickel, A. Peschke, M. Tyas, I. Mjör, S. Bayne, M. Peters, J.F. Rule, K. Van Dijken, N. Opdam, FDI World Dental Federation criteria for the clinical evaluation of dental restorative materials. *Dent. Mater.* 26, 103–104 (2010).
<https://doi.org/10.1016/j.dental.2009.11.005>
 9. M.A. Melo, L.N. Niu, Y.J. Chen, D.D. Arola, D.H. Pashley, F.R. Tay, Antibacterial and remineralizing effects of bioactive glass-containing composites. *Dent. Mater.* 36, 112–123 (2020). <https://doi.org/10.1016/j.dental.2019.10.012>
 10. J.W.V. van Dijken, P. Pallesen, Long-term dentin retention of etch-and-rinse and self-etch adhesives in Class I and II restorations. *Dent. Mater.* 24, 915–922 (2008).
<https://doi.org/10.1016/j.dental.2007.11.005>
 11. S. Imazato, R. Torii, Y. Tsuchitani, J.F. McCabe, S. Ebisu, Incorporation of antibacterial monomer MDPB into resin composite. *J. Dent. Res.* 80, 152–156 (2001).
<https://doi.org/10.1177/00220345010800020701>
 12. F. Li, J. Zhang, P.P. Chatterjee, Y. Zhang, M.D. Weir, H.H.K. Xu, Antibacterial and remineralizing effects of bioactive glass-containing nanocomposites. *Dent. Mater.* 29, 352–361 (2013). <https://doi.org/10.1016/j.dental.2012.12.005>
 13. N. Beyth, I. Yudovin-Farber, R. Bahir, A.J. Domb, E.I. Weiss, Antibacterial effects of quaternary ammonium methacrylate containing composites. *Dent. Mater.* 24, 1232–1238 (2008). <https://doi.org/10.1016/j.dental.2008.01.005>
 14. C.L. Davidson, A.J. de Gee, Adaptation of composite restorations to cavity walls. *J. Dent. Res.* 63, 104–107 (1984). <https://doi.org/10.1177/00220345840630020101>
 15. J.L. Ferracane, H. Condon, T. Okabe, P. Kettenum, K. DeRouen, A randomized clinical trial of posterior composites. *J. Dent. Res.* 77, 1539–1546 (1998).
<https://doi.org/10.1177/00220345980770070801>
 16. J.L. Ferracane, *Materials in Dentistry: Principles and Applications* (Lippincott Williams & Wilkins, Philadelphia, 2011)